

Ordinary differential Equations first order first degree

① Variable separation :- $f(x)dx = \varphi(y)dy$
 $\int f(x)dx = \int \varphi(y)dy + C$

② Homogeneous equation :-
Homogeneous function :- $f(x, y)$ is called
 Homogeneous function
 $f(cx, cy) = c^k f(x, y)$ (k) degree of Homogeneity
 $\rightarrow \frac{dy}{dx} = \frac{f(x, y)}{\varphi(x, y)}$, where $f(x, y)$ and $\varphi(x, y)$ are
 homogeneous functions of same degree.

Substitution

$$\begin{aligned} y &= vx \\ \rightarrow \frac{dy}{dx} &= v + x \frac{dv}{dx} \end{aligned}$$

③ ODE reducible to homogeneous equation

$$\frac{dy}{dx} = \frac{ax + by + c}{a'x + b'y + c'}$$

① Compute $ab' - ba'$
 if $ab' - ba' = 0$ / if $ab' - ba' \neq 0$

$$\text{if } ab' - ba' = 0$$

$$\frac{dy}{dx} = \frac{ax + by + c}{m(ax + by) + c'}$$

$$ax + by = v$$

$$a + b \frac{dy}{dx} = \frac{dv}{dx}$$

$$\text{if } ab - ba' \neq 0$$

$$x = X + h$$

$$y = Y + k$$

$$\begin{cases} ah + bk + c = 0 \\ a'h + b'k + c' = 0 \end{cases} \rightarrow \text{unique values of } h \text{ \& } k$$

④ Exact differential Equations :- Let $M(x,y)$ and $N(x,y)$ be functions of x and y . the Equation

$$Mdx + Ndy = 0$$

is called

$$\begin{aligned} d(f(x,y)) &= 0 \\ f(x,y) &= c \end{aligned}$$

Exact diff. eqn if $\exists$ a function $f(x,y)$ such

$$d(f(x,y)) = Mdx + Ndy$$

$$\frac{\partial f}{\partial x} dx + \frac{\partial f}{\partial y} dy$$

Solution

$$f(x,y) = c$$

Example :-

$$y^2 dx + 2xy dy = 0$$

$$\text{Take } f(x,y) = y^2 x$$

$$d(f(x,y)) = \frac{\partial f}{\partial x} dx + \frac{\partial f}{\partial y} dy = y^2 dx + 2xy dy$$

Solution

$$y^2 x = C$$

Thm:- The necessary and sufficient condition for a diff eqn $Mdx + Ndy = 0$ to be exact is

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$

Proof:- Let $Mdx + Ndy = 0$ be an exact diff eqn.
given $\exists f(x,y)$ s.t.

$$d(f(x,y)) = Mdx + Ndy$$

To prove :- $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$

$$\frac{\partial f}{\partial x} dx + \frac{\partial f}{\partial y} dy = Mdx + Ndy$$

we get

$$M = \frac{\partial f}{\partial x}$$
$$\frac{\partial M}{\partial y} = \frac{\partial^2 f}{\partial y \partial x}$$

$$N = \frac{\partial f}{\partial y}$$
$$\frac{\partial N}{\partial x} = \frac{\partial^2 f}{\partial x \partial y}$$

$$\Rightarrow \frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$

Conversely:- Given $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$

Aim • To find $f(x,y)$ s.t. $d(f(x,y)) = Mdx + Ndy$

Let $g(x,y) = \int M dx$ $\rightarrow$ Partial integral of M w.r.t x & constant

Let $g(x, y) = \int m \, dx$ → term of m w.r.t x keeping y constant

$$\frac{\partial g(x, y)}{\partial x} = m$$

Claim:- $\left(N - \frac{\partial g}{\partial y} \right)$ is function of y only

T.P. $\frac{\partial}{\partial x} \left(N - \frac{\partial g}{\partial y} \right) = 0$

$$\begin{aligned} \frac{\partial}{\partial x} \left(N - \frac{\partial g}{\partial y} \right) &= \frac{\partial N}{\partial x} - \frac{\partial^2 g}{\partial x \partial y} = \frac{\partial N}{\partial x} - \frac{\partial}{\partial y} \left(\frac{\partial g}{\partial x} \right) \\ &= \frac{\partial N}{\partial x} - \frac{\partial m}{\partial y} = 0 \end{aligned}$$

Take

$$f(x, y) = g(x, y) + \int \left(N - \frac{\partial g}{\partial y} \right) dy$$

Claim

$$d(f(x, y)) = m \, dx + N \, dy$$

Take those terms of function N , which does not contain x .

$$\begin{aligned} d(f(x, y)) &= d(g(x, y)) + N \, dy - \frac{\partial g}{\partial y} \, dy \\ &= \frac{\partial g}{\partial x} \, dx + \cancel{\frac{\partial g}{\partial y} \, dy} + N \, dy - \cancel{\frac{\partial g}{\partial y} \, dy} \\ &= m \, dx + N \, dy. \end{aligned}$$

Working Rule:-

$$m \, dx + N \, dy = 0$$

...

Working Rule :-

$$M dx + N dy = 0$$

- ① Identify M and N
- ② Compute $\frac{\partial M}{\partial y}$, $\frac{\partial N}{\partial x}$

some
↓
exact diff Eqn

Not
↓
Not exact diff Eqn.

solution is

$$\int M dx + \int (\text{terms in } N, \text{ not containing } x) dy = C$$

keeping y constant

Question :- Solve $(2x - y + 1) dx + (2y - x - 1) dy = 0$

Solution :-

$$M = 2x - y + 1 \quad N = 2y - x - 1$$
$$\frac{\partial M}{\partial y} = -1 \quad \frac{\partial N}{\partial x} = -1$$

Exact diff. Eqn

Solution is

$$\int (2x - y + 1) dx + \int (2y - 1) dy = C$$

y constant

$$(x^2 - xy + x) + y^2 - y = C$$

Question :-

$$y(2xy + e^x) dx = e^x dy$$

Solution :-

$$M = 2xy^2 + e^x y \quad N = -e^x$$
$$\frac{\partial M}{\partial y} = 4xy + e^x \quad \frac{\partial N}{\partial x} = -e^x$$

Solution:-

$$M = 2xy^2 + e^x y$$

$$\frac{\partial M}{\partial y} = 4xy + e^x$$

$$N = -e^x$$

$$\frac{\partial N}{\partial x} = -e^x$$

$$2xy^2 dx + ye^x dx - e^x dy = 0$$

$$\boxed{2x dx} + \boxed{\frac{e^x}{y} dx - \frac{e^x}{y^2} dy} = 0$$

$$M = \frac{e^x}{y}$$
$$\frac{\partial M}{\partial y} = -\frac{e^x}{y^2}$$

$$N = -\frac{e^x}{y^2}$$
$$\frac{\partial N}{\partial x} = -\frac{e^x}{y^2}$$

$$x^2 + \int \frac{e^x}{y} dx + \int 0 dy = C$$

Keeping y constant

$$\Rightarrow x^2 + \frac{e^x}{y} = C$$

* If an Eqn of the form $Mdx + Ndy = 0$ is not exact, then it can be made exact by multiplying with a function of x and y $\rightarrow$ Integrating factor

Not unique

No standard method to find I.F in all situations.

Rule I :- Guess

Rule II :- If the given equation $Mdx + Ndy = 0$ is not exact and $Mx + Ny \neq 0$

Rule II :- If the given equation is homogeneous and $Mx + Ny \neq 0$ then $\frac{1}{Mx + Ny}$ is an I.F.

* M and N are homogeneous functions of same degree.

Example 1:-

$$x^2 y dx - (x^3 - y^3) dy = 0$$

$$M = x^2 y$$

$$N = y^3 - x^3$$

$$\frac{\partial M}{\partial y} = x^2$$

$$\frac{\partial N}{\partial x} = -3x^2$$

Not Exact diff Eqn.

$$Mx + Ny = x^3 y - x^3 y + y^4 = y^4 \neq 0$$

$$\text{I.F.} = \frac{1}{y^4}$$

Eqn

$$\frac{x^2}{y^3} dx - \left(\frac{x^3}{y^4} - \frac{1}{y} \right) dy = 0$$

$$M = \frac{x^2}{y^3}$$

$$N = \frac{1}{y} - \frac{x^3}{y^4}$$

$$\frac{\partial M}{\partial y} = -\frac{3x^2}{y^4}$$

$$\frac{\partial N}{\partial x} = -\frac{3x^2}{y^4}$$

Solution

$$\int \frac{x^2}{y^3} dx + \int \frac{1}{y} dy = C$$

$y \text{ constant}$

$$\frac{x^3}{3y^3} + \ln|y| = C$$

question:-

$$x^2 y dx - (x^3 - y^3) dy = 0$$

$$\frac{x^2 y}{x^3 - y^3} = \frac{dy}{dx}$$

$$y = vx \quad , \quad \frac{dy}{dx} = v + x \frac{dv}{dx}$$

$$\frac{vx^3}{x^3 - v^3 x^3} = v + x \frac{dv}{dx}$$

$$\frac{v}{1-v^3} - v = x \frac{dv}{dx}$$

$$\frac{v^4}{1-v^3} = \frac{v - v + v^4}{1-v^3} = x \frac{dv}{dx}$$

$$\int \frac{dx}{x} = \frac{1-v^3}{v^4} dv = \int \frac{1}{v^4} - \frac{1}{v} dv$$

$$\ln|x| = \frac{v^{-4+1}}{-4+1} - \ln|v| + C$$

$$\ln|vx| = -\frac{1}{3} \cdot \frac{1}{v^3} + C$$

$$v = \frac{y}{x}$$

$$\ln|y| = -\frac{1}{3} \frac{x^3}{y^3} + C$$

$$\frac{1}{3} \frac{x^3}{y^3} + \ln|y| = C$$